

1. Let $(V, \|\cdot\|_V)$ be a Banach space. Let $f : [a, b] \rightarrow V$ be a continuous function and $\varepsilon > 0$. If f is differentiable in (a, b) and $\|f'(x) - f'(y)\|_V \leq \varepsilon$ for any $x, y \in (a, b)$, then prove that

$$\left\| \frac{f(b) - f(a)}{b - a} - f'(t) \right\|_V \leq \varepsilon \quad \forall t \in (a, b).$$

[Hint: Consider the function $\phi : [a, b] \rightarrow V$ defined by $\phi(s) := f(s) - f(a) - f'(t)(s - a)$ for every $s \in [a, b]$, to which we apply the Lagrange inequality.]

2. Let $(V, \|\cdot\|)$ be a normed space over the field $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$.
- (a) Define what is meant by the statement that $C \subset V$ is a non-empty convex subset of V .
 - (b) Show that the open ball $B(x_0, r)$ is a convex subset of V .
 - (c) Show that if C is a convex subset of V , then its closure $\overline{C}$ is also closed.
3. Let $(V, +, \cdot)$ be a real vector space.
- (a) Define what is meant by $C \subset V$ is a non-empty convex subset of V .
 - (b) Let $V := C([0, 1], \mathbb{R})$ be the space of continuous functions from $[0, 1]$ to $\mathbb{R}$. Consider the mapping $N : V \rightarrow \mathbb{R}$ defined by

$$N(f) := \sup_{x \in [0, 1]} e^{\arctan(1-x)} |f(x)|.$$

- (i) Show that N is a norm on V and that N is equivalent to the norm $\|\cdot\|_\infty$.
- (ii) Let

$$C := \{f \in V : \sin x \leq f(x) \leq 2x \quad \forall x \in [0, 1]\}$$

then, show that C is a **non-empty closed convex** subset of (V, N) .

4. Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give a counterexample.
- (a) Let $(V, +, \cdot)$ be a vector space over $\mathbb{R}$. If C_1 and C_2 are convex subsets of V with $C_1 \cap C_2 \neq \emptyset$, then each set $C_1 \cap C_2$, $C_1 \cup C_2$ and $C_1 + C_2$ is convex.
 - (b) Let A be a square matrix $n \times n$. Let $b \in \mathbb{R}^n$ and let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be defined by $f(x) := \phi(Ax + b)$ for some function $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^n$. If ϕ is convex, then the function f is also convex.
 - (c) If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, then also λf is convex for every $\lambda \in \mathbb{R}$.